EPRC Methods Workshop Series

Difference-in-Differences in Education Research

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LCDS Longitudinal Causal Data Science Lab

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Importance of Difference-in-Differences

- The availability of richer datasets and advances in computational power have transformed education research.
- A common goal of empirical DiD research is to identify the causal effects of educational policies and interventions on student outcomes.
- DiD is useful when random assignment to treatment is not feasible.

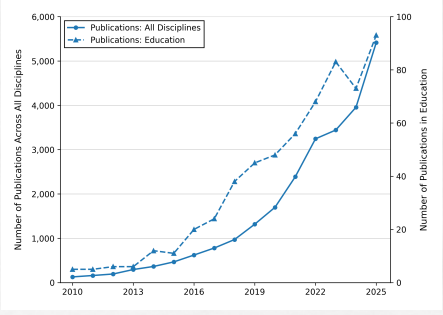


Figure 1. Annual Number of Publications, 2010–2025

Note. Annual publication counts were obtained from Web of Science using the following search query: “difference-in-difference” OR “difference in difference” OR “difference-in-differences” OR “difference in differences” OR “differences-in-differences” OR “differences in differences”.

Popularity of Difference-in-Differences Methods

Currie et al. (2020)

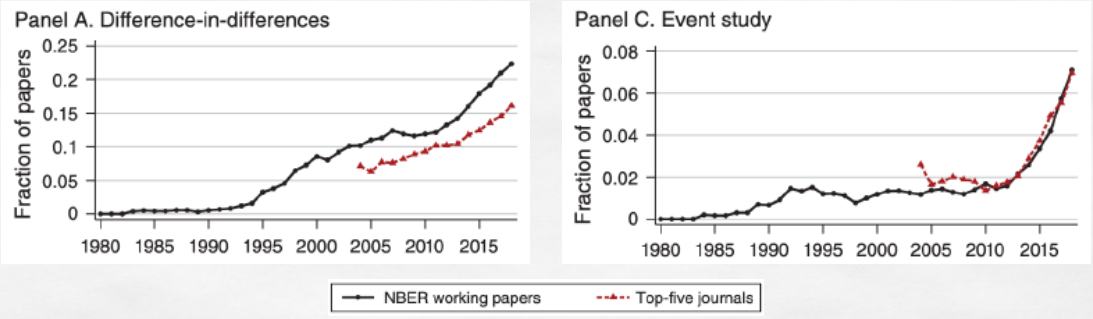


Figure 2. Quasi-Experimental Methods in Applied Microeconomics Papers

DiD Example: Gifted and Talented Programs

Research question: Does participation in New York City's Gifted and Talented (G&T) program improve student attendance and academic achievement?

Design: Longitudinal student-level data and staggered DiD, following students who entered G&T in Grades 1–3.

Main findings:

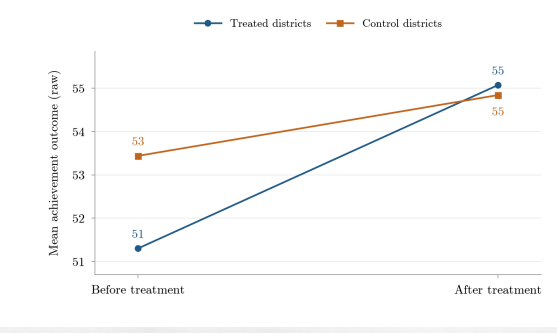
- Modest, sustained improvements in attendance and academic achievement, including middle-school mathematics and English Language Arts proficiency.
- Benefits accumulated over time, with effects reaching 0.11–0.18 standard deviations for students entering in Grade 3.

Strickland, K., Chan, W., & Gottfried, M. A. (2026). *Longitudinal Effects of NYC's Gifted and Talented Program: Evidence From a Staggered Difference-in-Differences Approach.* Educational Evaluation and Policy Analysis. doi:10.3102/01623737261438146

The Classic 2×2 DiD Design

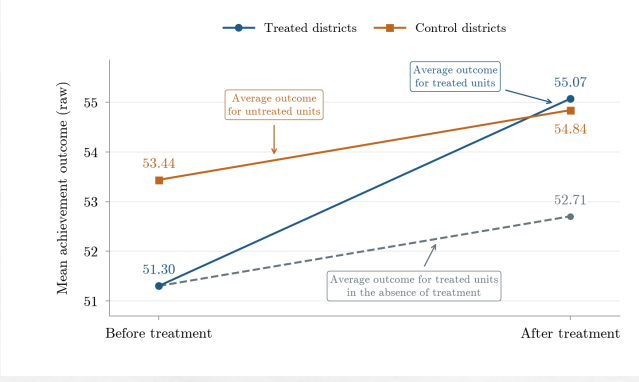
Let's start with the simplest DiD setup: **two groups and two time periods.**

- **Two groups:** Treatment and comparison
- **Two periods:** Before and after treatment



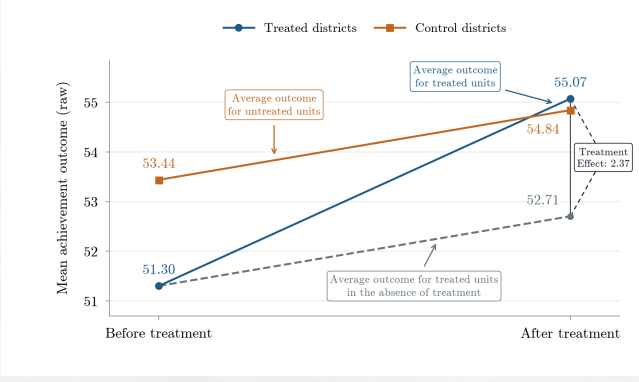
The Parallel Trends Assumption

In the absence of treatment, the treated and control groups would have experienced the same average changes in outcomes over time.



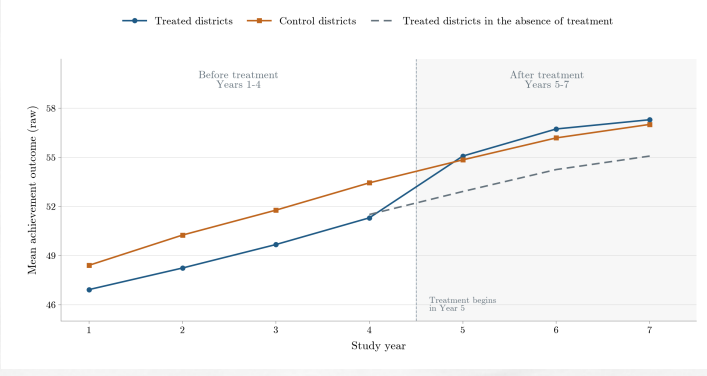
The Parallel Trends Assumption

In the absence of treatment, the treated and control groups would have experienced the same average changes in outcomes over time.



DiD with Multiple Periods

The standard 2 $\times$ 2 DiD design can be extended to multiple periods.



The TWFE Model

TWFE is widely used to estimate treatment effects.

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}$$

Model Piece	Interpretation
α_i	Unit fixed effects
λ_t	Time fixed effects
D_{it}	Treatment indicator, equal to 1 if unit i is treated at time t, and 0 otherwise
β	Treatment effect
u_{it}	Error term

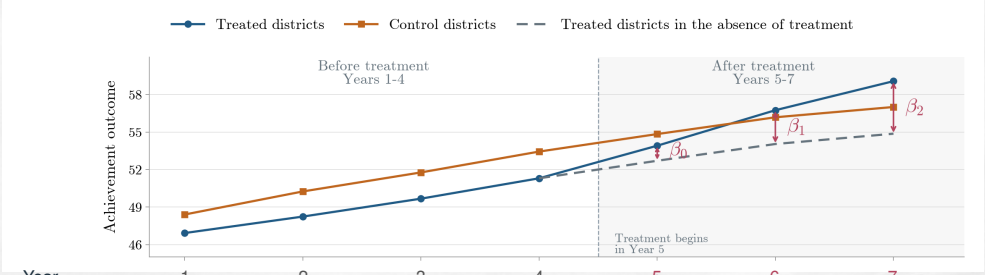
The coefficient from the TWFE model equals

$$\beta = \left(\overline{Y}_{G,\text{post}} - \overline{Y}_{G,\text{pre}}\right) - \left(\overline{Y}_{C,\text{post}} - \overline{Y}_{C,\text{pre}}\right)$$

TWFE event study

Event-study models estimate separate coefficients for periods before and after treatment.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}\{t - g = k\} + u_{it}$$



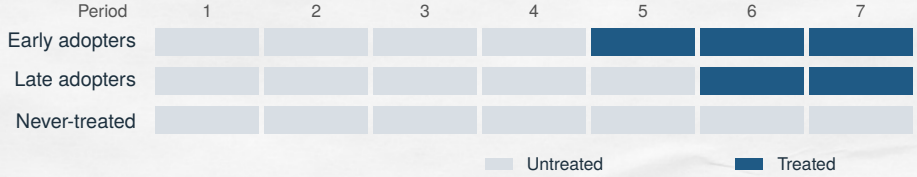
Treatment begins in Year 5, so $g = 5$.
Treated outcomes in Years 5–7 are illustrative.

Staggered DiD

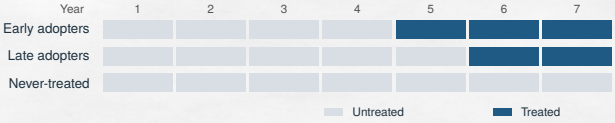
Units enter treatment at different times.

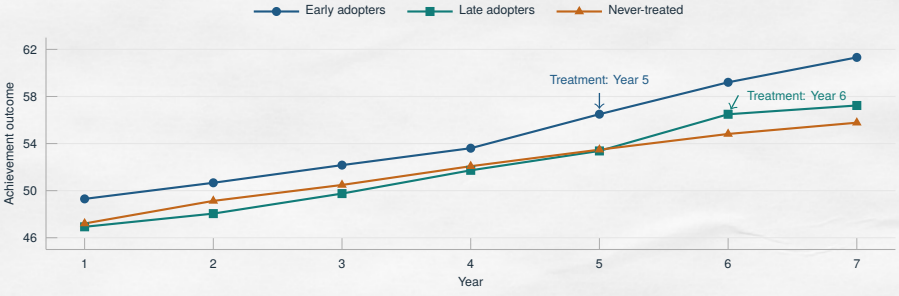
Staggered DiD designs are widely used in empirical research.

- Baker et al. (2022) document that **54.7%** of DiD studies in leading finance and accounting journals (2000–2019) use staggered DiD designs.



Staggered DiD: Treatment Timing and Outcomes





TWFE Can Be Biased in Staggered DiD

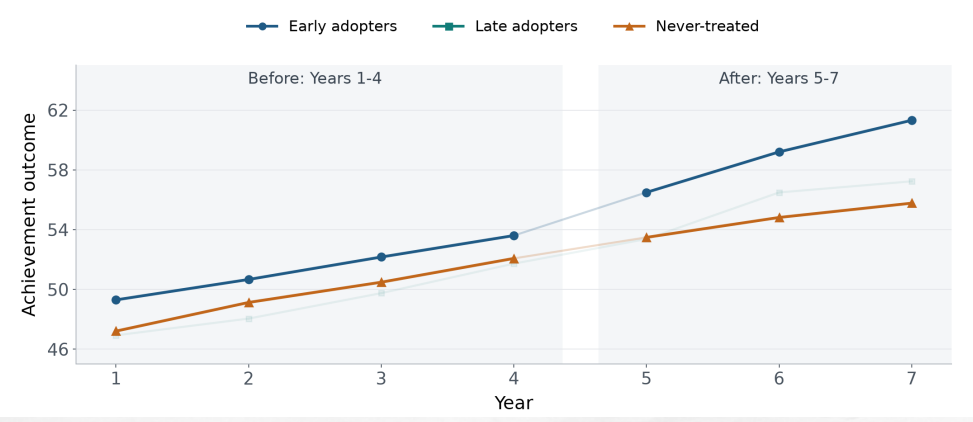
The 2WFE estimator combines multiple 2×2 DiDs (Goodman-Bacon, 2021).

More precisely, it combines four types of comparisons:

1. Early adopters to never treated
2. Late adopters to never treated
3. Early adopters to late adopters
4. Late adopters to early adopters

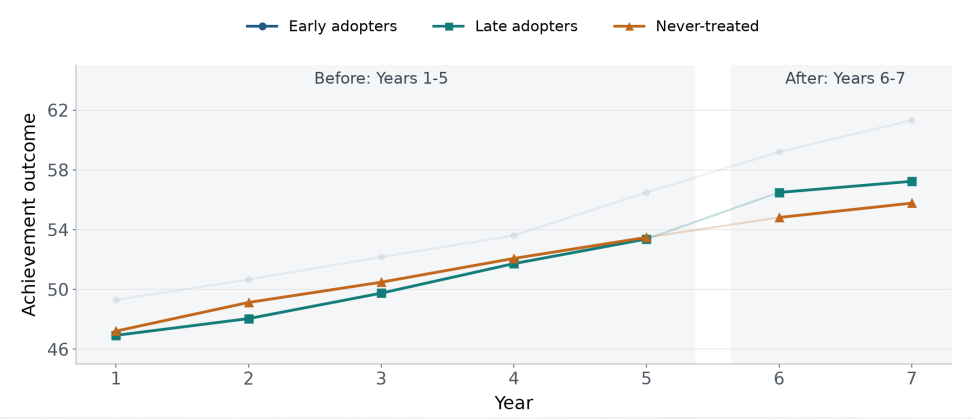
Number 4 is problematic: the adopters are already treated!

1. Early adopters vs. never-treated



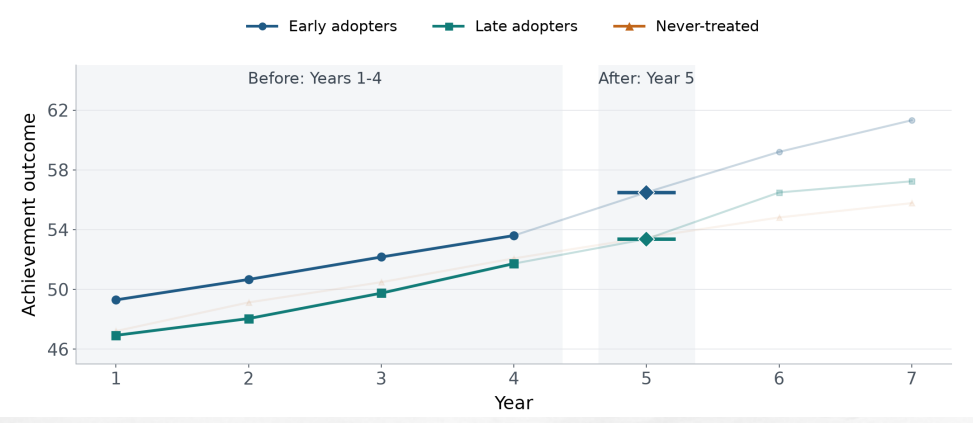
Comparison group is untreated in both selected windows.

2. Late adopters vs. never-treated



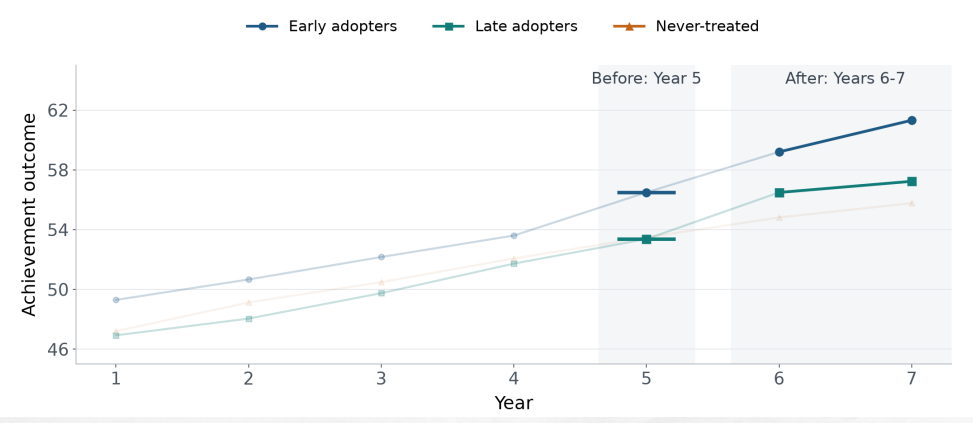
Comparison group is untreated in both selected windows.

3. Early adopters vs. late adopters



Comparison group is untreated in both selected windows.

4. Late adopters vs. early adopters



Early adopters are already treated in both selected windows.

Staggered DiD Estimator

Callaway and Sant'Anna (2021) propose an alternative approach:

- Estimate group-time average treatment effects, $\text{ATT}(g, t)$.
- Use never-treated or not-yet-treated units as control groups.
- Aggregate group-time effects to obtain overall or dynamic treatment effects.

$$\text{ATT}(g,t) = \left(\overline{Y}_{g,t} - \overline{Y}_{g,g-1}\right) - \left(\overline{Y}_{C,t} - \overline{Y}_{C,g-1}\right)$$

g	First treatment year, which identifies the treated cohort.
t	Year in which the outcome is measured.
C	Control group: never-treated units or units not yet treated by year t.

Group-Time Estimates: A Numerical Example

Control group: Never-treated districts. Outcome: Achievement (raw scale).

Year	Early adopters ($g = 5$)	Late adopters ($g = 6$)
5	$\widehat{\text{ATT}}(5,5) = 1.48$	
6	$\widehat{\text{ATT}}(5,6) = 2.86$	$\widehat{\text{ATT}}(6,6) = 1.78$
7	$\widehat{\text{ATT}}(5,7) = 4.02$	$\widehat{\text{ATT}}(6,7) = 1.57$

Post-treatment ATTs use Year $g - 1$ as the baseline: Year 4 for early adopters and Year 5 for late adopters.

Example, early adopters in Year 7:

$$\widehat{\text{ATT}}(5,7) = (61.32 - 53.61) - (55.77 - 52.08) \approx 4.02$$

Weighted Averages: Summarizing Treatment Effects

After estimating $\text{ATT}(g,t)$, we can aggregate treatment effects to answer different research questions.

Method & Research Question	How Effects Are Averaged
Simple ATT What is the overall average treatment effect?	Average all post-treatment $\text{ATT}(g,t)$ values. Cohorts with more units or more post-treatment periods receive greater weight.
Group ATT How do treatment effects differ across cohorts?	First average treatment effects over time within each cohort. Then combine the cohort averages, weighting by the number of units in each cohort.
Dynamic ATT How do treatment effects evolve over time since treatment?	Average effects across cohorts at the same event time ($e = t - g$), weighted by cohort size.
Calendar ATT How do treatment effects vary across calendar periods?	Average effects across treated cohorts in the same calendar period, weighted by cohort size.

Statistical Inference

Work in Progress

Alternative Methods for Staggered DiD

Beyond Callaway and Sant'Anna (2021), several alternative methods have been developed for staggered DiD settings.

Method	Reference
Interaction-Weighted Event Study	Sun & Abraham (2021)
Imputation Estimator	Borusyak et al. (2024)
Two-Stage DiD	Gardner (2022)
Extended TWFE (ETWFE)	Wooldridge (2025)
Alternative DiD Estimators	de Chaisemartin & D'Haultfoeuille (2022)
Stacked DiD	Wing et al. (2024)

These methods will be discussed in more detail in our reading group.

Wooldridge - Pooled OLS and ETWFE

Work in Progress